Collapse Around the Circuit: Induction Heads Survive Bounded Model Collapse While Their Errors Are Inherited

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Abstract

We study model collapse inside a mechanism we can read: the induction circuit of Singh et al. (2024). A two-layer attention-only transformer learns an in-context symbol task and is retrained for up to 20 generations on pools that mix a fraction λ of real data with data sampled from its predecessor. In the main sweep the circuit survives: test accuracy never falls below 0.979 and ablating layer 1 costs 0.66–0.76 accuracy in every generation. What degrades is the model's free generative choice. The probability it places on symbols absent from the context follows a two-parameter inheritance law, $m_{g+1} = a(1-\lambda)m_g + \varepsilon$ ($a = 0.951$, $R^2 = 0.994$), which predicts held-out real-data fractions. Removing those off-context samples from the synthetic pools cuts the gain a to 0.19, and training-data attribution finds them 12–23$\times$ over-represented among the most influential examples: the errors are carried by specific synthetic samples, not by a damaged circuit. A pre-registered extension of the law to higher sampling temperatures failed four of five tests. Collapse ran away only without real data, and the circuit then broke once correct labels in the training pool fell to chance level.

Keywords: model collapse; induction heads; in-context learning

Introduction

Language models adapt to patterns shown only in their prompt, and much of this in-context learning (ICL) is attributed to *induction heads*: a previous-token head copies each token's predecessor into its position, and an induction head one layer later attends from the current token to an earlier match and copies what followed it (Olsson et al., 2022). Singh et al. (2024) isolate this circuit in a minimal symbol-copying task. Separately, models trained recursively on generated data suffer *model collapse*, losing the tails of the data distribution (Shumailov et al., 2023, 2024); collapse is avoided when enough real data remains (Bertrand et al., 2023) or when data is accumulated rather than replaced (Gerstgrasser et al., 2024). These lines of work have not met. We ask whether recursive training damages the ICL circuit itself or only what the circuit produces, and we make a quantitative, pre-registered prediction about when collapse runs away.

Background Attention patterns suggest a circuit but are not causal evidence, so we follow Conmy et al. (2023) and ablate heads. To trace collapse back to specific training examples we use gradient-based data attribution (Pruthi et al., 2020), in the spirit of Chen et al. (2026).

Methods

Task and model. Each sequence gives $K{=}4$ symbols (from 16) a random label (from 6). A study block lists the pairs and a query block repeats the symbols in an independent order; the model predicts each label, which only content-based matching can solve. Unlike Singh et al. (2024) (two pairs, image symbols), we use tokens and query every symbol in shuffled order, so no positional shortcut works. The *base* model predicts only the query labels. The *extended* model then freely generates one more in-context symbol ($sq2$, uniform over the four in real data) and its label; $sq2$ is the degree of freedom that can collapse. The model has 2 attention-only layers of 4 heads ($d{=}64$, learned and rotary positions, tied embeddings, cosine learning-rate decay; 35k parameters). Hyper-parameters were chosen on validation loss and confirmed once on a held-out test seed.

Collapse pipeline. Generation 0 trains on a 20k-sequence real pool. Generation g trains a *fresh* model on a 20k pool with a fraction λ of real sequences and the rest sampled from generation $g{-}1$ at temperature $T{=}1$. Prompts stay real and all answers are model-generated (*prompt-real* regime). Each pool is split 90/10 into train and validation; test data is fresh real data. A generation must pass a strict gate (validation accuracy and a causal layer-1 check) or is retrained, which happened 20 times in 680 main-sweep generations with no final failures. We sweep $\lambda \in \{0, 0.1, 0.25, 0.5, 1\}$ for 12 generations, with 10 seeds for the extended model (20 generations at $\lambda{=}0$) and 5 for the base model.

Metrics. *Off-context mass*, the probability $p(sq2 \mid \text{context})$ places on symbols not in the context, measures hallucination; query NLL on real data measures confidence. The circuit is measured causally, as the accuracy lost when every layer-1 head is mean-ablated (at most about 0.83, down to chance at 1/6). This is agnostic to how the circuit is implemented, which matters because 5% of generations use a non-canonical variant that attention probes misread. We fit the inheritance law on $\lambda \in \{0, 0.25, 1\}$ and test it on $\{0.1, 0.5\}$. Interventions remove off-context or wrong-label samples from the synthetic pools, accumulate pools across generations, change the pool size, inject a one-off 5% label-noise pulse, and vary depth, width and task size.

Pre-registration. From one calibration seed, the law was fitted per temperature $T \in \{0.7, 1.3, 1.6\}$. It predicts a critical fraction $\lambda^* = 1 - 1/a(T)$ whenever $a(T) > 1$. Five pass/fail tests were committed to git before the confirmatory runs (5 seeds per cell).

Results

The circuit survives; its outputs do not. At generation 0 (test accuracy 0.998) a layer-0 head sends 0.79 of each label's attention to the preceding symbol and a layer-1 head 0.93 of each query's attention to its match. Accuracy then stays $\geq$0.979 in every generation, and the accuracy lost by ablating layer 1 is 0.71 at generation 0 and 0.69–0.74 at the last generation for every λ (Figure 1c). Confidence erodes instead: without real data, query NLL rises from 0.006 to 0.030 by generation 19 while accuracy is 0.996 (Figure 1b). The base model, with no free choice, shows only this effect: at $\lambda{=}0$ its accuracy stays 1.000 while query NLL rises from 0.0002 to 0.0058 by generation 11.

An inheritance law. Off-context mass grows from 0.013 to 0.138 by generation 19 without real data, and to 0.057, 0.035, 0.021 and 0.012 by generation 11 at $\lambda = 0.1, 0.25, 0.5$ and 1. The law fits all 630 transitions with $a = 0.951$ (95%

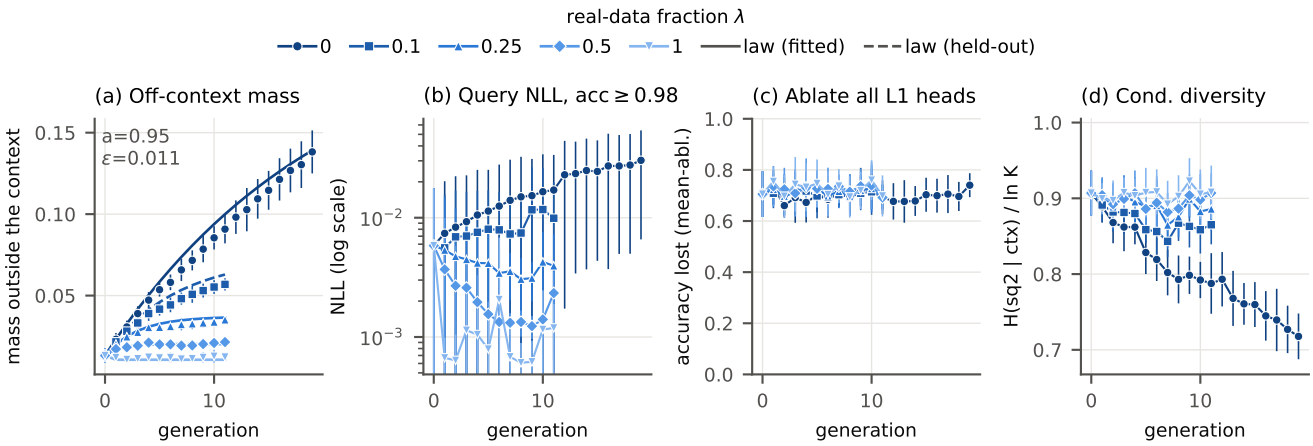


Figure 1: Main sweep (10 seeds, mean ± sd). (a) Off-context mass with the inheritance law; solid lines are fitted fractions, dashed lines are held-out predictions. (b) Query NLL rises without real data while accuracy stays ≥0.98. (c) Accuracy lost when all layer-1 heads are mean-ablated stays near 0.7: the circuit survives. (d) Conditional diversity of $sq2$ falls only without real data.

CI 0.942–0.959), $\varepsilon = 0.0105$ and $R^2 = 0.994$. Fitted without $\lambda \in \{0.1, 0.5\}$, it predicts them with RMSE 0.003, within one seed SD at 68% of generations (Figure 1a). It overshoots the plateau: for $\lambda{=}0$ it predicts $m^* = 0.21$, while mass is 0.13 at generations 16–19 and still rising. The gain holds across 3- and 4-layer, MLP and wider models ($a = 0.94$–1.04, $R^2 \geq 0.98$) and is lower on larger tasks (0.84 at $K{=}6$, $R^2 = 0.80$; 0.85 with 32 symbols); a one-layer model cannot learn the task (accuracy 0.64).

Specific synthetic samples carry the errors. Removing off-context samples from each synthetic pool cuts the gain from 0.95 to 0.19 (95% CI -0.02–0.43), while removing wrong labels leaves it at 0.95 (0.93–0.97). Attribution agrees: off-context samples are 12–23× over-represented among the top 1% most influential examples. Accumulating pools lowers the gain to 0.62 (0.52–0.71), and a 5k pool raises it to 1.03 (1.02–1.04, vs 0.91 for matched 20k runs), so mass runs away to 0.58 without real data. A label-noise pulse decays at the predicted rate with real data (half-life 2.2 vs 2.1 generations at $\lambda{=}0.25$; 0.97 vs 0.93 at 0.5) but far more slowly without it (33 vs 14).

Table 1: Pre-registered temperature tests (confirmatory seeds 0–4). Only the qualitative prediction (test 5) held.

Test	Result	Observed ($T = 0.7, 1, 1.3, 1.6$)
1. $a(T)$ rises with T	fail	$a = -0.04, 0.93, 1.14, 1.04$
2. Inside 95% band	fail	3 of 12 cells
3. Contracting cells slow	fail	2 of 3 cells at $T{=}0.7$ (gain ≈ 0)
4. Runaway iff $a(1{-}\lambda) \geq 1$	fail	runaway only at $\lambda{=}0$; no seed in the 6 cells with $\lambda > 0$
5. Higher T: more mass, less homogenisation	pass	mass 0.011, 0.079, 0.77, 0.82; entropy 0.40, 0.78, 0.91, 0.93

The pre-registered prediction failed. Only the qualitative trade-off held (Table 1, test 5); the predicted critical fractions ($\lambda^* \approx 0.30$ at $T{=}1.3$, 0.52 at 1.6) do not exist: mass exceeded 0.5 in every seed at $\lambda{=}0$, yet no seed ran away in the six cells with 20–50% real data predicted to.

When collapse runs away, the circuit breaks. In those runs the share of correct labels in the synthetic pool fell to 0.15 ($T{=}1.3$) and 0.05 ($T{=}1.6$) by generation 9, but the circuit outlasted the labels. At $T{=}1.6$ it was intact at generation 4 with 26% correct labels (accuracy 0.93, layer-1 effect 0.67) and gone at generation 5 with 15% (0.37 and 0.06); at $T{=}1.3$ it broke at the same share (0.46 and 0.20). Both breaks came as correct labels fell to chance (1/6).

Discussion

In the prompt-real regime, bounded collapse leaves the induction circuit intact: every self-generated query is still answered by copying from a real context, which re-teaches the same computation. Collapse concentrates in the free choice, mainly as off-context hallucination inherited through the off-context samples. The circuit is lost only when copying stops paying, once correct answers are no more common than chance. The failed pre-registration bounds the linear account: at high temperature real data damps errors far more than a constant gain allows, so the law describes collapse locally but does not locate tipping points (its single-seed calibration also underestimated the gain at $T{=}1$: 0.86 vs 0.95). Survival also depends on supervising only the answers: with every token supervised (full language modelling) and no real data, the layer-1 effect falls from 0.83 to 0.71 and accuracy to 0.88 by generation 9, even with real prompts. Other limits: the 35k-parameter model, 16 symbols, 5–10 seeds and at most 20 generations. Two lessons carry over: accuracy can hide eroding confidence, and a modest share of real data, or accumulating data, sharply limits what a model inherits.

Contribution Statements

Group members: Sunay Master (2677874), Matthew Purkiss (2697424), Harshil Vallabh (2656158)

Project: *Interpretability of Model Collapse* (working title: *Collapse around the circuit: induction heads under recursive training*)

Research question. A small transformer learns the induction-head symbol task of Singh et al. (2024) and is then trained recursively on data generated by its previous generation (Shumailov et al., 2024). Does collapse degrade the induction circuit itself, or only what the circuit is asked to produce? How does the fraction of real data mixed into each generation change this, in both the base and the extended version of the task?

Division of work

Each member contributes one third (33.3%) of the project.

Member	Share	Experiments	Writing and admin
Sunay	33.3%	Core implementation and collapse pipeline; A+C sweeps and hyper-parameter selection; temperature test and poison-pulse runs	Method section; project outline and change plan
Matthew	33.3%	Single-generation analysis: training curves, attention maps, embeddings, convergence and loss floor; extra logging; headline figure	Introduction, Background and Results sections; extended abstract LaTeX; Faculty of Science AI ethics statement
Harshil	33.3%	Inheritance-law fit; gate audit and statistics; finite-data control run; mechanism analysis of the hallucination	Discussion and limitations; NeurIPS 2024 checklist; README.txt and requirements.txt

Sunay Master

- Build the study→query task generator (base and extended), the 2-layer transformer, the training loop, and the collapse pipeline (real/synthetic mixing, convergence gate with retries, collision-free seeding).
- Build the circuit probes (previous-token and induction scores, causal head ablation), conditional diversity and off-context mass; run the full sweep over real-data fractions, seeds and both modes; select hyper-parameters.
- Run the pre-registered temperature test of the critical real-data fraction and the poison-pulse experiment, and write the Method section.

Matthew Purkiss

- Produce the single-generation training curves, attention-map and embedding figures, and check that extended mode converges and reaches its predicted loss floor.
- Add validation loss and all-query perplexity to the logging, add mean-ablation to the probes, and build the composite headline figure.
- Write the extended abstract (Moodle template) including Introduction, Background and Results, and the Faculty of Science AI ethics statement.

Harshil Vallabh

- Fit the recursion model to the saved results, with held-out fractions as a test, and audit which generations fail a stricter convergence gate.
- Run the control that gives generation 0 the same data budget as later generations, and analyse which heads produce the off-context (hallucinated) probability.
- Write the Discussion and limitations, the NeurIPS 2024 checklist, README.txt and requirements.txt, and check that every number in the abstract traces to a file.

Shared by all members

- The two-page extended abstract (PDF), the full code, and checking every reference.
- The final contribution statement, which will report any changes to this plan and why.

Agreement

All group members have read and agree to this division of work.

Sunay Master Matthew Purkiss Harshil Vallabh

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   Justification: The Discussion states that the circuit result holds only with real prompts and answer-only supervision (it degrades under full language-model training), and lists the 35k-parameter model, 16 symbols, 5–10 seeds and at most 20 generations. Results report where the inheritance law fails: it overshoots the $\lambda{=}0$ plateau and failed its pre-registered extension to other temperatures.

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Justification: The code, the raw outputs of all 311 runs and every analysis file are submitted with the paper and kept at `https://github.com/sunay777/NLP_Project1`. All data is synthetic and generated by `src/data.py`. `README.txt` gives one command per experiment and analysis, `requirements.txt` pins the packages (Python 3.12–3.13), and `python -m experiments.abstract_numbers` regenerates every number in the paper from the raw runs. Model weights are not included; they are rebuilt by rerunning the jobs.

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Justification: Methods state the splits (each pool split 90/10 into train and validation; test data is fresh real data), that hyperparameters were chosen on validation loss and confirmed once on a held-out test seed, and the gate that retrains unconverged generations. The optimizer and exact values are in `src/config.py`, and the tuning results in `results/tuning_*.json`.

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   Justification: All runs used CPUs only, one thread per run, on two Apple-silicon Macs; no GPU or cluster. With 9 runs in parallel, one extended generation took about 5 minutes per core. The 311 runs trained 3,293 generations (plus retries), about 330 CPU core-hours in total. Matthew’s share (200 runs) took about 17 hours of wall time on 9 parallel workers; Sunay’s share (111 runs, including the larger models) took about 26 hours on 9 workers over two sessions, about 5 hours of which were 4-layer runs repeated after a fix to the convergence gate. Earlier work not in the paper (a first sweep, hyper-parameter tuning, pilots and the temperature calibration) used additional compute.

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